MSIF: Multisize Inference Fusion-Based False Alarm Elimination for Ship Detection in Large-Scale SAR Images

Chao Zhang, Chule Yang, Kaihui Cheng, Naiyang Guan, Hongbin Dong, and Baosong Deng

***Abstract*— Ship detection in large-scale synthetic aperture radar (SAR) images has essential value in both military and civilian applications. However, due to the complexity of the background and the simplicity of the texture, ship detection in large-scale SAR images is prone to false alarms, such as similar-shaped reefs, islands, sea clutter, and inland buildings. This article proposes a multisize inference fusion framework to eliminate false alarms and improve the overall performance of ship detection in large-scale SAR images. In this framework, a multisize slicer is proposed to expand the scale range of image expression. Then, a detection model library is built to keep various types of models for different task scenarios and requirements. Finally, two subapproaches are proposed for false alarm elimination, namely, pixel feature filtering (FAE-pff) and multisource fusion (FAE-msf), to reduce false detection results in the output of the detection model. FAE-pff calculates how obvious each target is relative to the background and eliminates less obvious results. FAE-msf obtains bounding boxes and corresponding confidences from multiple inference sources and fuses them through weighting and updating them to achieve complementation and enhancement of information. Various experiments were conducted to evaluate the performance of each module qualitatively and quantitatively. It proves the effectiveness of the proposed framework, which can achieve more correct detections while greatly reducing erroneous detections.**

***Index Terms*— False alarm elimination, information fusion, large-scale SAR image processing, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor that can work in all-days and all-weather conditions. This imaging mechanism makes it widely used in military and civilian fields. Target detection using SAR images is a hot research topic in SAR applications, and ship detection is one of the most demanding and practical technologies [1], [2].

Manuscript received November 17, 2021; revised February 14, 2022; accepted March 8, 2022. Date of publication March 11, 2022; date of current version April 12, 2022. This work was supported by the National Natural Science Foundation of China under Grant 62006242. *(Corresponding author: Chule Yang.)*

Chao Zhang and Hongbin Dong are with the Faculty of Computer Science and Technology, Harbin Engineering University, Harbin 150001, China (e-mail: zhang_chao@hrbeu.edu.cn; donghongbin@hrbeu.edu.cn).

Chule Yang, Kaihui Cheng, Naiyang Guan, and Baosong Deng are with the Defense Innovation Institute (DII), Academy of Military Sciences, Beijing 100071, China (e-mail: yangchule@126.com; chengkaihui1999@126.com; nyguan@sina.com; dbs310@163.com).

Digital Object Identifier 10.1109/TGRS.2022.3159035

Traditional SAR ship detection methods mainly focus on manually designing features to detect ships, such as ship wakes, textures, and scale [3], [4]. However, these methods have low detection efficiency and do not have the adaptability of the scene, and the design of features needs to be modified according to the transformation of the scene. False alarms are prone to occur during detection in coastal areas. In addition, speckle noise and ship motion blur make some targets lose their inherent features, which brings difficulties to traditional ship detection methods. Most current research on a wide range of ship detection methods is based on constant false alarm rate (CFAR) detection technology [5]–[8]. To determine potential targets in the ocean, Ao *et al.* [5] used the generalized gamma distribution clutter model to construct an efficient multiscale CFAR detector. For ship detection in large-scale SAR images, Li *et al.* [6] developed a superpixel-level two-stage CFAR technology. Traditional algorithms usually provide higher detection performance for only specific scenarios, and their flexibility is insufficient. In addition, these CFAR-based technologies rely heavily on the accuracy of clutter statistical models. The background contamination will cause inaccurate parameter calculations, leading to seriously missed inspections.

With the development of deep learning technology, target detection based on deep learning is becoming more mature. Many researchers apply deep learning target detection methods to ship detection in SAR images. Especially in recent years, an end-to-end network has been proposed so that the feature does not need to be manually designed, enhancing the model's generalization ability. Since the proposal of AlexNet [9] and its success in large-scale image detection, deep learning methods have been widely used in computer vision tasks, such as image segmentation and target detection. In the field of target detection, deep learning target detection methods are mainly divided into anchor-based and anchor-free. Anchor-based development started earlier, mainly including R-CNN [10], Fast R-CNN [11], and other two-stage representative methods; the main feature of this type of method is that the predicted position of the bounding box is accurate. In addition to the one-stage method represented by SSD [12] and YOLO9000 [13], this type of method is mainly characterized by fast detection speed. Later, because the design of using anchor is relatively inflexible for target detection, the idea of using key points to replace the anchor frame came into being. For example,

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

CornerNet [14] and CenterNet [15] are all representative methods of anchor-free, which both improve the detection precision while reducing the number of model calculations.

In the earlier, the SAR image ship detection method is based on deep learning: crop the patch from the entire SAR image and then use CNN to identify whether the cropped patch belongs to the target object. These methods cannot overcome the impact of detection precision caused by changes in ship shape and size. Later, with the development of target detection technology under deep learning, Ren *et al.* [16] and Kang *et al.* [17] proposed a series of anchor-based target detection frameworks and solved this problem by using region-based CNN to fuse features from different levels. Jiao *et al.* [18] used a multiscale neural network to solve the multiscale and multiscene SAR ship problem. Recently, with the proposal of anchor-free technology, because anchor-free target detection technology has inherent design advantages for target detection with a large-scale span, this happens to be the problem of targets in large-scale SAR images. Therefore, many researchers focus on applying anchor-free frame target detection technology to target detection under SAR images. For example, the feature balance and refinement network (FBR-Net) proposed in [19] is used for the detection of multiscale ship targets in complex scenes. Cui *et al.* [20] proposed that the spatial shuffle group enhancement (SSE) attention module combined with the anchor-free frame CenterNet has a good performance on large-scale SAR images. However, compared with the anchor-based target detection model, the precision of the anchor-free target detection model is lower.

In large-scale SAR image ship detection, on the one hand, the size of the ship relative to the whole picture is small, so there is less information mapped to the feature map, which will significantly reduce the performance of ship detection. On the other hand, large-scale SAR images are rich in scenes, which can easily cause misreporting of ships, such as clutter on land, islands, reefs, and ocean waves. Moreover, the rich scenes of ships will also bring difficulties to detection, such as ships docking and multiple ships berthing side by side. In order to solve this problem, Pang *et al.* [21] use a simple patch classification network to filter out areas without targets. In addition, Wang *et al.* [22] proposed a fast and straightforward preclassification method for sea block selection based on bias classification. The above two methods both filter out the possible patches of the target by adding a terrain classification network. Although the second method reduces the calculation amount of the screening network, it is still not simple enough. Furthermore, Cui *et al.* [20] also proposed an SSE attention module for large-scale SAR images. The detection range is focused on the area where the target may appear through the attention mechanism, and good performance is achieved in combination with CenterNet. Moreover, there are fewer large-scale SAR image datasets, which is also a challenge for large-scale SAR image detection. Therefore, Niu *et al.* [23] proposed an optical image-based SAR image generation method to solve this problem.

Based on the above analysis, this article proposes a multisize inference fusion (MSIF) framework to eliminate false alarms and improve the overall performance for ship detection in large-scale SAR images. The framework comprises three modules: a slicer module, a detection module, and a false alarm elimination module. The slicer module proposes to slice large-scale SAR images in multiple sizes to reduce computational complexity, prepare for feature scaling, and maintain a complete target presentation. By inputting slice images of different sizes into the detection module, the target feature can be scaled to a more significant degree. The module also maintains a model library containing various types of algorithms, which can be adaptively applied to different task scenarios and requirements. In the false alarm elimination module, two subapproaches are proposed in a progressive structure: pixel feature filtering (FAE-pff) and multisource fusion (FAE-msf). FAE-pff calculates the difference between the foreground and the background of each bounding box to distinguish the correct detection from the false alarm. FAE-msf fuses the detection results from different sources and eliminates the false alarms by weighted NMS and belief rethresholding.

In the following contents, Section II introduces the proposed framework in detail. Section III conducts experiments on the public dataset to analyze the performance of the framework. Section IV gives conclusions and future work.

II. Methodology

The overall framework of ship detection in large-scale SAR images is shown in Fig. 1. The framework mainly includes three processes: preprocessing, detection, and postprocessing. The slicer module is proposed as a preprocessing method to slice large-scale images in different sizes. The detection module maintains a model library, which can be deployed flexibly according to different task requirements. In the postprocessing process, two false alarm elimination methods are proposed to reduce erroneous detection results from two perspectives: pixel feature filtering (FAE-pff) and multisource fusion (FAE-msf).

A. Slicer

The raw SAR image generally has the characteristics of large scale and high resolution. Taking into account application requirements, such as real-time and mobile terminal deployment, directly inputting large-scale original images into the detection model not only has low accuracy but may also be unable to calculate due to computing resource constraints. Therefore, the image needs to be cropped before detection. For these reasons, this article proposes a multisize slicer that can set different slice sizes to cut large-scale SAR images.

Second, a detection model trained with a specific image size usually only has effective detection capabilities for targets within a specific scale. In order to provide higher quality data for the detection module, the multisize slice method proposed in this article can further scale the target features through image scaling in the subsequent detection model and scale the features that are not initially in the detection range of the model to this scope. Therefore, selecting slice images of different sizes and inputting them into the detection model will produce different detection results. On the one hand, this design can generate more true positive detection and improve

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

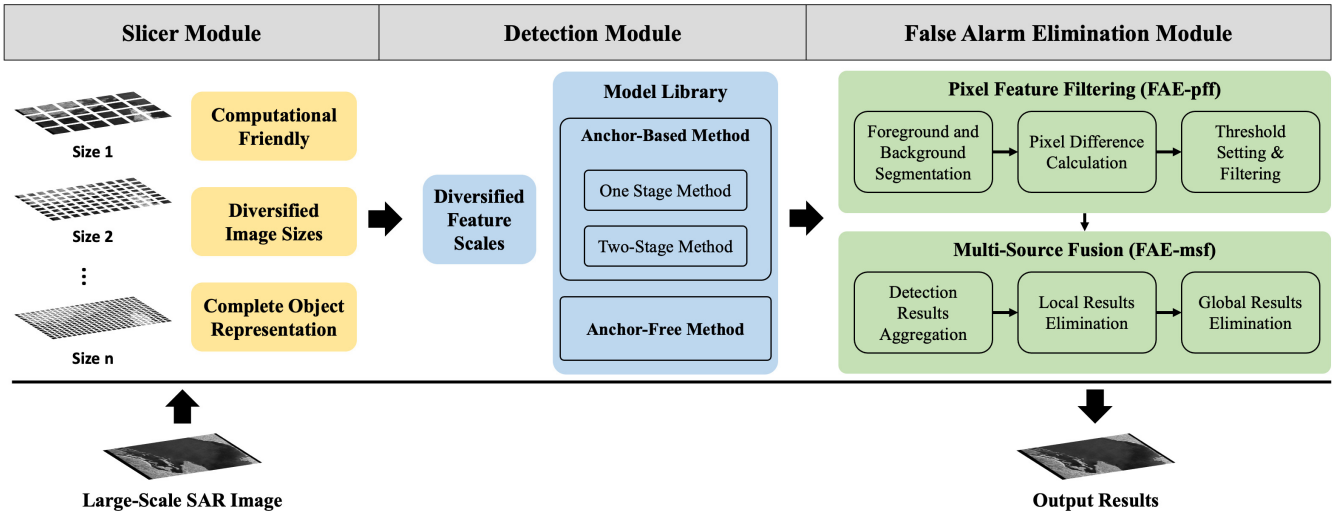


Fig. 1. Overall flowchart of the MSIF method. Three modules are proposed, namely, slicer, detection, and false alarm elimination. The slicer module cuts the raw image with different slice sizes. The detection module contains various types of models for different scenarios and requirements. The false alarm elimination module includes two subapproaches considering the pixel feature of each detection and fusion of detection results from multiple sources.

the recall rate, and on the other hand, it can reduce the requirement for multiscale training of the model.

The setting of the slice size s_i follows the following formula:

$$\forall s_a \nmid \forall s_b [s_a, s_b \in s_i, s_i \in N^*]. \tag{1}$$

For any two-slice sizes s_a and s_b, they are recommended not to be set to be divisible by each other. Therefore, it can retain a complete presentation of the target in at least one size. In addition, when the slice size cannot be evenly divided by the size of the large-scale image, the part beyond the boundary is filled with black pixels.

B. Detection

The detection module extracts features from each sliced image and locates the position of ships. Most of the existing CNN-based networks use feature pyramid networks (FPNs) or path aggregation network (PANet) structures in the neck to solve multiscale detection problems. However, it is still challenging to detect objects of very different sizes, such as ships. By feeding the output images from the slicer module into the backbone, the ship's features can be scaled once before entering the neck, which will result in more true positive detection results.

This module also maintains a model library for different task requirements. According to the latest classification of object detection methods, end-to-end detection models can be categorized into anchor-based and anchor-free methods, and anchor-based methods can be further categorized into one- and two-stage methods. Different methods have their strengths and weaknesses, selected and modified according to different task scenarios and needs.

Depending on whether to use anchors or not, object detection methods are usually categorized into two classes: anchor-based and anchor-free. For the former class, anchor-based methods typically use many anchors to predict the bounding box in advance. They can effectively improve the ability to detect objects, but dense proposal regions also consume many computing resources. Therefore, anchor-based methods are used for tasks that require high detection accuracy, especially recall rate, but have less stringent requirements for real-time performance. For the latter class, anchor-free methods directly use the features extracted by the network to predict the class and location, and their network structures are usually simple and computationally cheap. However, although such methods can alleviate the imbalance between positive and negative proposals, they cannot distinguish overlapping targets and blurred semantic boundaries. As a result, anchor-free methods are more suitable for scenarios where the distribution of objects is not dense, and they are easier to be deployed on mobile platforms with limited computing resources.

According to the classification and bounding box regression steps, object detection methods are divided into one- and two-stage two types. The two-stage method first generates many candidate bounding boxes and then classifies and fine-tunes each bounding box. It is characterized by higher accuracy but generally slower speed. The one-stage method returns the category confidence and position of the object simultaneously and is characterized by faster speed. Therefore, generally speaking, the one-stage method is more suitable for real-time applications, such as video stream detection, while the two-stage method is more suitable for detailed analysis, such as still image detection.

C. Pixel Feature Filtering for False Alarm Elimination (FAE-pff)

After the detection module, through the analysis of the correct detection results and the erroneous detection results, it can be found that the ship targets are mainly distributed in the inshore and offshore areas, and the erroneous detection targets are mainly composed of offshore islands, and reefs and inland objects that look like ships. Further analyzing the pixel distribution of the types mentioned above of targets, as shown

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

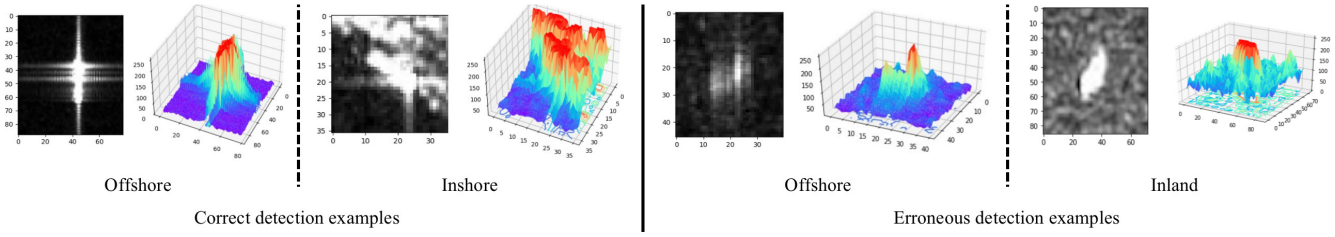


Fig. 2. Pixel feature distribution of correct detections and false alarms. As shown on the left, correct detection shall be in offshore and inshore, which has a larger difference between foreground and background. Erroneous detection can be either in offshore (islands and reefs) or inland (ship-like objects), which has a smaller difference between foreground and background.

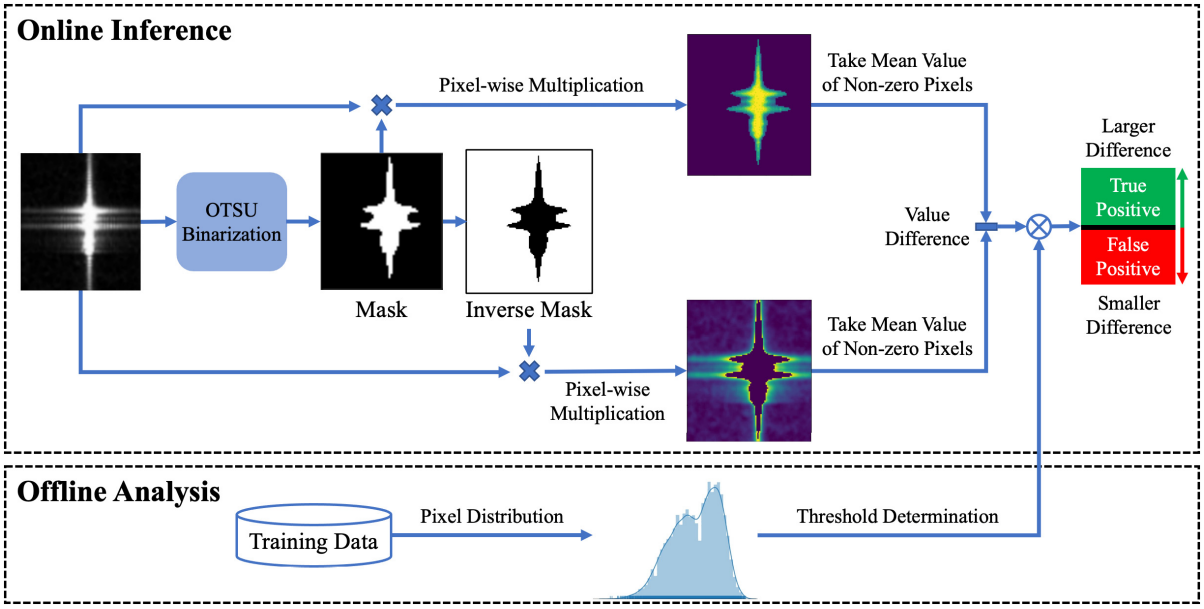


Fig. 3. Flowchart of FAE-pff. For the online process, it first calculates the best segmentation of the pixels in each bounding box through the OTSU algorithm and obtains the masks of both foreground and background. Then, the average pixel value of the foreground and background is calculated by multiplying the image by the corresponding mask and then subtracted to get the difference between them. For the off-line process, a threshold is set according to the distribution of pixel differences in the training set. The detection result with a difference less than the threshold is regarded as a false positive detection.

in Fig. 2, it can be found that the imaging principle of SAR images follows the reflection mechanism of radar waves. Radar waves reflect strongly on metal structures, are weak on rock-like structures, and are weakest on water. Corresponding to SAR images, the pixel value of the ship area is the highest, the pixel value of the island and land is low, and the pixel value of the water surface is the lowest. From this, it can be found that there is a significant pixel difference between the target area and the background area of the correct detection result, while the difference is slight in the erroneous detection result. Therefore, the false alarm can be eliminated by describing the difference.

This article proposes an FAE-pff approach to eliminate false alarms according to the pixel feature within each detected bounding box based on the above analysis. The workflow of this approach is shown in Fig. 3. The pixels in the bounding box contain two types of semantics (foreground and background). First, use the OTSU algorithm to find the threshold t that can best separate the foreground pixels from the background pixels. Traverse the pixel value range (0–255), and find a value t such that the difference between the mean value of the pixel value higher than the t value and the mean value of the pixel value lower than t value is the largest. It can be considered that this value t is the best segmentation value between the foreground and the background.

The value t is used to binarize the image within the bounding box. After normalization, the mask of the foreground area of the image can be obtained, and vice versa, the mask of the background area of the image can be obtained. Multiply the image in the bounding box by the mask of the foreground area to obtain the image of the foreground area, and calculate the average value of the image pixels of the foreground area. In the same way, find the average of the pixels in the background area. By subtracting the background pixels' average value from the foreground pixels' average value, the pixel difference between the foreground and the background can be obtained. The process of calculating the pixel difference can be expressed as follows:

$$P_{\text{dif}} = A \times \frac{\hat{A}}{255} - A \times \left[-\left(\frac{\hat{A}}{255} - I \right) \right] \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

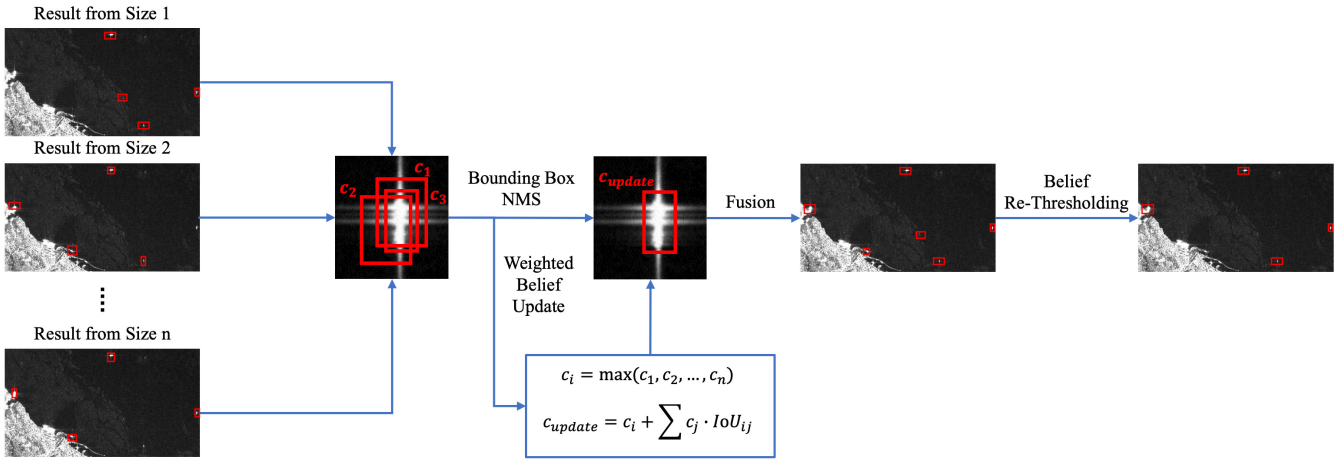


Fig. 4. Flowchart of FAE-msf. First, the detection results under various slice sizes are aggregated by transforming from each local coordinate into a global coordinate. According to the principle of NMS, the bounding box with the highest confidence is reserved for each target, and the confidence of the box is updated at the same time based on weighted NMS. Finally, referring to the confidence threshold set in the detection model, another value higher than that threshold is set to be a final judgment belief, and the ship whose belief is less than the value will be eliminated.

where the distribution of image pixel values in the bounding box can be represented by a matrix A, and then, the matrix $\hat{A}$ is the result of binarizing the image with the value of t. I is the identity matrix.

The range of pixel difference is determined by analyzing the distribution of pixel difference in the training set offline. Set the pixel difference threshold according to the value range to eliminate false alarms.

D. Multisource Fusion for False Alarm Elimination (FAE-msf)

After image slices of different sizes pass the detection model, the output detection results are, respectively, passed through the FAE-pff method to eliminate false alarms. Here, it is necessary to merge the detection results of slices with different sizes and infer all the detected targets. The proposal of this method is inspired by the traditional nonmaximum suppression (NMS) method [24], and on this basis, a weighted NMS algorithm is proposed. A multisource fusion inference method is proposed based on the principle that the confidence of the same target detected in multiple slices of different sizes is greater than the confidence of the target detected in a single size slice. The specific process is shown in Fig. 4. Integrate the detection results of multiple large and small slices, remove the bounding box of repeated detection targets according to the weighted NMS algorithm, update the confidence, and, finally, set the confidence threshold to eliminate false alarms.

The steps of removing redundant bounding boxes in the weighted NMS algorithm are the same as those of the NMS. First, select a confidence threshold, and filter out the confidence lower than the confidence. Then, select the bounding box with the highest confidence and calculate the IOU value with the remaining boxes. Delete the box with the IOU value higher than the threshold, multiply the calculated IOU value of the deleted box and the box with the highest confidence by the confidence of the deleted box, and then add it to the box with the highest confidence. Repeat the above operation until the end. The update of confidence can be expressed as follows:

$$c_{\text{update}} = c_i + \sum c_j \cdot \text{IOU}_{ij} \tag{3}$$

where c_i is the highest confidence and c_j is the confidence of the bounding box that overlaps the bounding box with the highest confidence. IOU_{ij} is the IOU value of the bounding box j and the bounding box i.

As a result, the updated confidence can effectively distinguish high-confidence targets detected multiple times in slices of various sizes and low-confidence targets detected in only one slice of size. Therefore, by adjusting and setting the confidence threshold, it is possible to effectively eliminate accidental targets with low confidence detected in a single size. The confidence update strategy is shown in (4), which maintains the same interval as the original confidence (threshold in the detection model). Since lower raw confidence may leave more false positives in the detection stage, a larger increase should be given in rethresholding, whereas higher raw confidence gets a smaller increase

$$T_{\text{update}} = T_{\text{origin}}{}^{(1+T_{\text{origin}})^{-1}}. \tag{4}$$

III. Experiments

Experiments were performed on the large-scale SAR ship detection dataset (LS-SSDD) to evaluate the proposed method. The experimental part is divided into the following parts. First, the experimental dataset, evaluation criteria, and detailed settings are introduced. Then, verify the fusion of multiscale detection results to achieve the enhancement and complementation of information. In addition, ablation experiments were performed on the proposed framework, and three representative models of the detection model were replaced to verify the function of each module. Finally, the method proposed in this article is compared with other methods, showing excellent performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

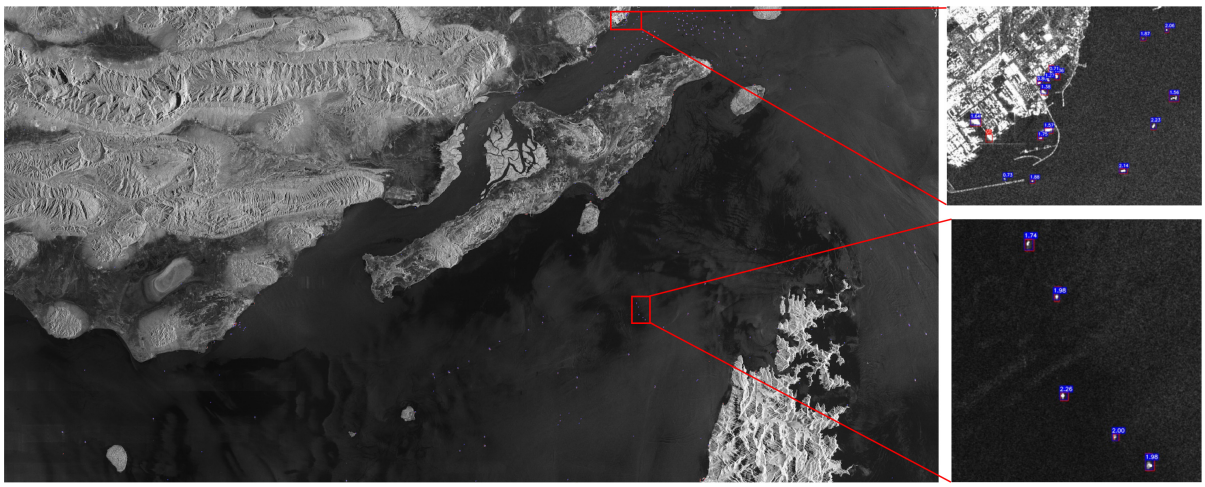


Fig. 5. Ship detection results in large-scale SAR image. Select representative inshore and offshore areas to zoomed-in view on the details, and display the detection results.

TABLE I
MULTISCALE DETECTION FUSION EXPERIMENT

Model	Input size	Prec(%)	Recl(%)	$F_1(\%)$	$F_2(\%)$	$AP_{50}(\%)$	TP	FP
416	600	82.39	66.13	73.37	68.84	60.95	3978	850
	800	89.07	69.51	78.08	72.70	67.40	4181	513
	1000	85.59	57.87	69.05	61.88	54.12	3481	586
	Fusion	82.16	72.57	77.07	**74.30**	68.85	**4365**	948
608	600	88.08	71.01	78.63	73.87	67.06	4271	578
	800	91.19	74.36	81.92	77.20	72.08	4473	432
	1000	88.49	65.45	75.25	69.05	62.00	4161	609
	Fusion	85.64	78.15	81.73	**79.54**	75.22	**4701**	788
800	600	91.00	67.93	77.79	71.56	65.48	4086	404
	800	94.30	69.33	79.91	73.21	68.20	4170	252
	1000	92.62	61.10	73.63	65.56	59.09	3675	293
	Fusion	88.48	74.21	80.72	**76.68**	72.57	**4464**	581

A. Experiment Preparation

1) Dataset: The LS-SSDD dataset [25] contains 15 large-scale SAR images, each with a size of 24 000 × 16 000. The scenes in the figure are vibrant, including land, ocean, and island. In order to facilitate the training of the model, the large image is divided into 9000 subimages with a size of 800 × 800 as the training set. The images are annotated with the support of the automatic identification system (AIS) and Google Earth.

2) Evaluation Metrics: This article uses precision, recall, F1 score, and average precision (AP) to evaluate the performance of the detection model. Among them, precision and recall are two evaluation metrics that are widely used in detection models. They are defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

where TP is a positive class predicted as a positive class, FP is a negative class predicted as a positive class, and FN is a positive class predicted as a negative class.

As (5) shows that the precision rate and the recall rate affect each other, so the detection model usually uses the F_1 score and AP to evaluate the overall performance of the model. Since (5) indicate that precision and recall affect each other, the detection model usually uses AP and F_1 to evaluate the overall performance of the model. F_1 is an indicator for balanced evaluation of precision and recall, which is extended to a more extensive definition of F_β scores. Among them, F_2 and $F_{0.5}$ scores are commonly used in statistics, and in F_2 scores, the recall rate is more important than the precision rate. The $F_{0.5}$ score precision rate is more important than the recall rate.

F_β is defined as follows:

$$F_\beta = \left(1 + \beta^2\right) \cdot \frac{\text{precision} \cdot \text{recall}}{\left(\beta^2 \cdot \text{precision}\right) + \text{recall}}. \tag{6}$$

AP is the lower area precision–recall curve, and AP (mAP) is the average of multiple classes of AP. There is only one precision of ship detection in this article, so mAP = AP. AP_{50} represents the AP at which the detection is considered

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ABLATION STUDY OF THE SYSTEM

Type	Setting	Prec(%)	Recl(%)	$F_1(\%)$	$F_{0.5}(\%)$	$AP_{50}(\%)$	TP	FP
Anchor-Based (One Stage)	YOLOv5s (multi-scale)	73.56	83.39	78.16	75.34	79.72	5016	1803
	YOLOv5s + FAE-pff	76.86	82.34	79.51	77.90	79.04	4953	1491
	YOLOv5s + FAE-pff + FAE-msf	80.09	81.18	**80.63**	**80.31**	78.13	4883	**1214**
Anchor-Based (Two Stages)	Faster R-CNN (multi-scale)	75.61	38.04	50.61	63.14	33.88	2288	738
	Faster R-CNN + FAE-pff	76.11	38.04	50.73	63.42	33.90	2288	718
	Faster R-CNN + FAE-pff + FAE-msf	76.28	38.02	**50.75**	**63.50**	33.89	2287	**711**
Anchor-Free	YOLOx (multi-scale)	56.33	71.12	62.87	58.77	64.04	4278	3317
	YOLOx + FAE-pff	63.00	70.07	66.35	64.30	63.69	4215	2475
	YOLOx + FAE-pff + FAE-msf	78.46	64.06	**75.08**	**77.33**	59.38	3853	**1058**

confirmed when the Intersection over Union (IoU) between the detected bounding box and the ground truth is higher than 0.5. The definition of AP is given as follows:

$$\mathrm{AP} = \int_0^1 P(R)dR \tag{7}$$

where P and R represent the single-point value of precision and recall.

3) Settings: The software environment is used to implement the experiment: PyTorch 1.7, CUDA 11.1, and CUDNN 8.1.1. Hardware environment: CPU is Intel Core i9-10900k, and GPU is GeForce RTX 3090. Since only 2032 subimages in the 9000 subimages of the dataset contain ship targets, the remaining subimages are all scene images without ship targets. Therefore, 1859 subimages were randomly selected for training and 173 subimages for testing and verification. YOLOv5s (yolov5s is a lightweight version of yolov5, which uses the smallest network depth and feature map width), YOLOx, and Faster R-CNN are used as the representative algorithms of the three detection models deployed in the detection module.

B. Experiment Description

In this part of the experiment, a multisize detection fusion experiment will be performed first to verify that the detection results of different size slice input detection model fusion can achieve the effect of information enhancement and complementation. The second is to conduct ablation experiments to verify the phased effect of the detection module and the approach of eliminating false alarms. At the same time, three different types of detection models are deployed to verify the universality and effectiveness of the framework proposed in this article for various detection models. Finally, by comparing the method proposed in this article with the most advanced method, this article's advanced nature is verified.

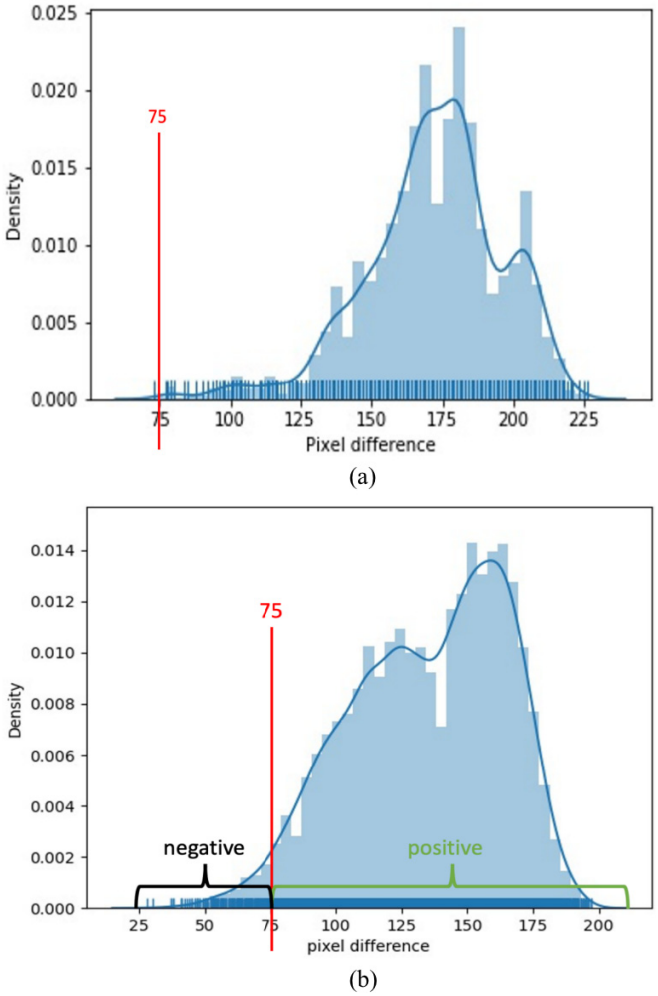


Fig. 6. Threshold setting of FAE-pff in the LS-SSDD dataset. (a) Threshold determination by statistical analysis of the pixel difference distribution in the training set. (b) Threshold application to the test set. Positive implies those bounding boxes that are regarded as correct detection results, while negative implies those false alarms, and they are eliminated.

C. Experiment Results

1) Result Visualization: Fig. 5 shows the ship detection results of the proposed method on large-scale SAR images. The two detailed pictures show the detection results of the inshore and the offshore, respectively. As shown in the figure,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

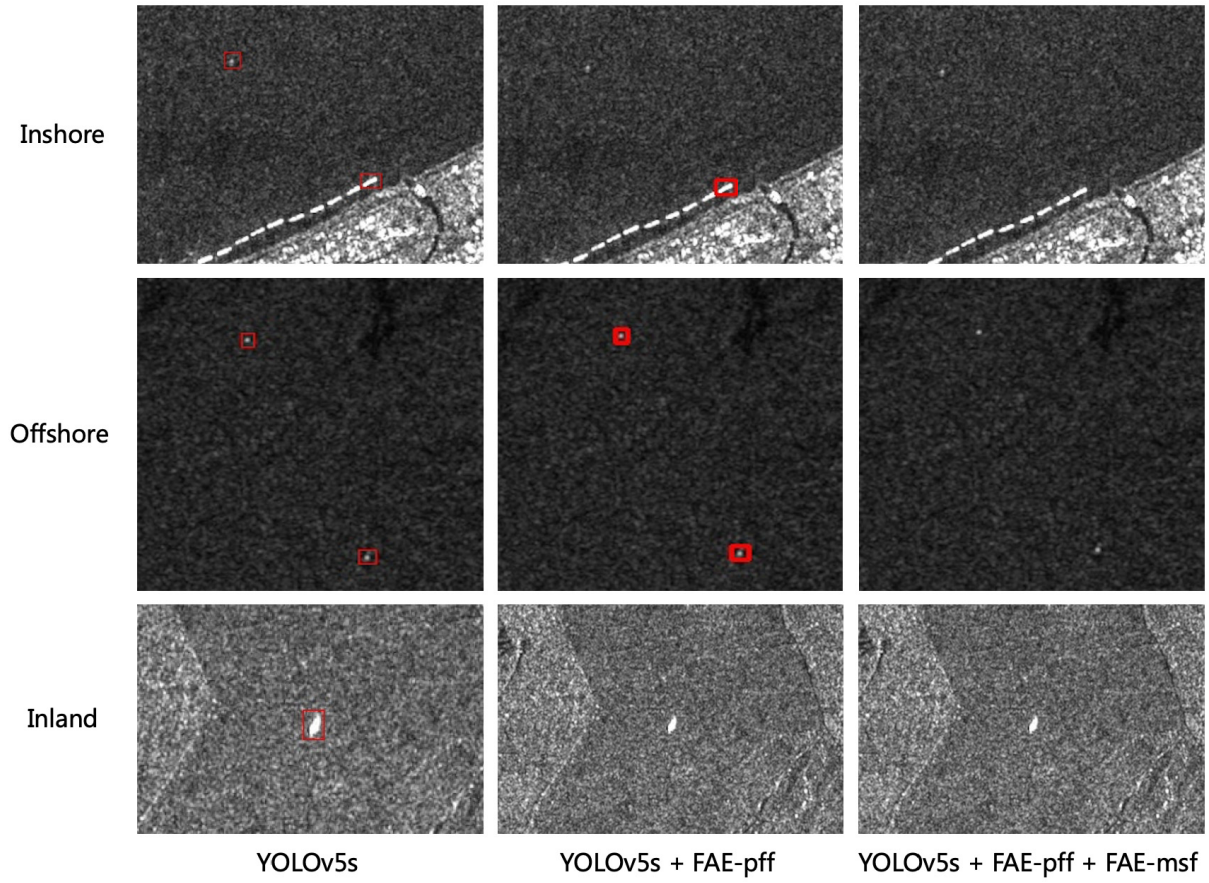


Fig. 7. Visualize the results of ablation experiments. Three false alarm samples in different scenarios (inshore, offshore, and inland) are selected to show the elimination effects of the two false alarm elimination approaches.

the method proposed in this article has achieved good results in both scenarios.

2) Multisize Inference Result: In order to verify the fusion of multisize detection results, which can achieve complementary information, the following experiments are designed. Using YOLOv5s as the detection model, the images in the training set of the LS-SSDD dataset were resized to three image sizes of 416, 608, and 800 to train models of three different scales. Then, the image slices of three sizes of 600, 800, and 1000 are input into the model for testing in turn as the test set. Finally, the detection results of the three input sizes are fused. The fusion method is to superimpose the detection results of three input sizes, then perform NMS processing, and retain the union of the three input detection results of different sizes.

The experimental results are shown in Table I. The fusion of multisize detection results can bring about a substantial increase in TP, which is also reflected in the changes in the value of F_2. Although FP also increases simultaneously, the increase in TP is greater than the increase in FP, and FP will be eliminated in subsequent steps. This result verifies that the fusion of multisize detection results realizes the complementation of information.

3) Threshold Setting: In the FAE-pff, it needs to set a threshold to achieve the eliminate effect and then how to determine the threshold to set, as shown in Fig. 6; first, statistically analyze the pixel difference distribution of the samples in the training set. It can be found that the pixel differences of the samples are basically above 75. Therefore, the pixel difference threshold is set to 75 in the test set. The detection results with pixel differences less than 75 are regarded as negative samples for elimination. The detection results with pixel differences more significant than 75 are regarded as positive samples for retention. Subsequent experimental results proved that the test result index after the negative samples were excluded from the comprehensive evaluation was higher than that of the nonexcluded samples.

4) Ablation Study: In this experiment, three latest and representative detection models were selected. These three models represent three different types of detection models (anchor-based one-stage model: YOLOv5s, anchor-based two-stage: Faster R-CNN, and anchor-free: YOLOx). Deploy them in the model library of the detection module for ablation experiments. According to the structure of the method proposed in this article, the multiscale detection results are merged first. Then, two false alarm elimination methods are sequentially added for testing. It can be seen from Table II that, for each detection model, after adding two false alarm elimination methods, in turn, the precision value gradually increases, and the value of F_1 gradually increases. Although the recall value will drop to a certain extent, the increase in F_1 and $F_{0.5}$ indicates that the decrease in TP value is much

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON WITH OTHER METHODS ON THE LS-SSDD DATASET

Method	Precision(%)	Recall(%)	$F_1(\%)$	$AP_{50}(\%)$
Faster R-CNN [26]	73.74	77.71	75.67	74.80
PANET [27]	74.91	76.32	74.92	73.33
Cascade R-CNN [28]	84.09	72.67	77.96	70.88
Libra R-CNN [29]	73.55	76.70	75.09	73.68
Grid R-CNN [30]	79.02	75.23	77.08	72.28
DCN [31]	74.13	76.87	75.48	73.84
EfficientDet [32]	62.14	67.49	64.70	61.35
Guided Anchoring [33]	87.94	70.86	78.48	69.02
HR-SDNet [34]	84.96	70.56	77.09	68.80
SSD-300 [12]	85.50	28.26	42.48	25.37
SSD-512 [12]	90.47	42.30	57.65	40.60
YOLOv3 [35]	87.64	26.24	40.38	24.63
RetinaNet [36]	94.62	55.51	69.97	54.31
GHM [37]	84.69	70.73	77.08	68.52
FCOS [38]	93.92	51.30	66.36	50.33
ATSS [39]	89.75	31.29	46.40	30.10
FreeAnchor [40]	55.30	77.67	64.60	71.04
FoveaBox [41]	**95.75**	53.03	68.26	52.32
MSIF (Ours)	86.58	**78.17**	**82.13**	**75.57**

smaller than the decrease in FP value, which proves the two false alarm approaches is conducive to the improvement of the overall framework performance. It also proves that the two approaches of eliminating false alarms have a gradual effect. At the same time, the experimental results under three different models can prove the universality of the false alarm elimination approaches proposed in this article to the detection model.

As shown in Fig. 7, the representative detection results of the YOLOv5s ablation experiment are selected for visual display, and it can be found that the two false alarm elimination approaches can eliminate the erroneous detection results of the YOLOv5s model very well. As shown in Fig. 7, the erroneous detection results in three different scenarios are selected (inshore, offshore, and inland). It can be found that FAE-pff can eliminate the erroneous detection results in inshore and inland very well. However, in the FAE-pff stage, part of the erroneous detection results in offshore, and the erroneous detection results in inshore could not be eliminated; both were eliminated in the FAE-msf stage, which can reflect the progressiveness and complementary of FAE-pff and FAE-msf approaches.

5) Comparison With Other Methods: In order to verify the performance of the method proposed in this article, different method tests were performed on the LS-SSDD dataset, and the performance indicators of the method proposed in this article were compared with other methods, as shown in Table III. It can be found that, in addition to precision, other indicators have reached the highest. Among them, F_1 reached 82.13%, an increase in nearly 4% from the second-highest cascade R-CNN's F_1. Compared with the AP_{50} value of faster R-CNN, the second highest, AP_{50}, has increased by nearly 0.8%. Therefore, it can be determined that the comprehensive performance of the method proposed in this article is in a leading position. It can also be found that the precision of the method proposed in this article is also higher than that of most methods. Moreover, those methods with higher precision are at the expense of the recall value, so it can be considered that the precision value is not worthy of comparison.

IV. CONCLUSION

An MSIF framework based on two false alarm elimination approaches is proposed in this article to solve the ship detection problem of false alarm suppression in ship detection in large-scale SAR images. The pixel feature filtering (FAE-pff) approach eliminates false alarm samples by measuring the pixel distribution features of correct detection results and false detection results. The multisource fusion (FAE-msf) approach is to reevaluate the fusion detection result's confidence by fusing the detection result of the multisize slice to achieve the effect of suppressing false alarms.

The experiment proves that the comprehensive performance index of this method is in a leading position. The method proposed in this article has better overall detection performance, and both false alarm approaches can effectively reduce false alarms. In future work, we will pay more attention to ship detection in more complex scenarios, including nearshore ship detection.

REFERENCES

[1] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.

[2] M. Tello, C. Lopez-Martinez, and J. J. Mallorqui, "A novel algorithm for ship detection in SAR imagery based on the wavelet transform," *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 2, pp. 201–205, Apr. 2005.

[3] L. M. Kaplan, "Improved SAR target detection via extended fractal features," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 436–451, Apr. 2001.

[4] A. C. Copeland, G. Ravichandran, and M. M. Trivedi, "Localized radon transform-based detection of ship wakes in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 1, pp. 35–45, Jan. 1995.

[5] W. Ao, F. Xu, Y. Li, and H. Wang, "Detection and discrimination of ship targets in complex background from spaceborne ALOS-2 SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 2, pp. 536–550, Feb. 2018.

[6] T. Li, Z. Liu, R. Xie, and L. Ran, "An improved superpixel-level CFAR detection method for ship targets in high-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 184–194, Jan. 2018.

[7] L. Zhang, S. Lu, D. Xiang, and Y. Su, "SAR image ship detection based on a superpixel non-window fast cfar strategy," in *Proc. SAR Big Data Era*, 2021, pp. 1–4.

[8] T. Zhang, Z. Yang, H. Gan, D. Xiang, S. Zhu, and J. Yang, "PolSAR ship detection using the joint polarimetric information," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8225–8241, Nov. 2020.

[9] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 25, Dec. 2012, pp. 1097–1105.

[10] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 580–587.

[11] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[12] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 21–37.

[13] J. Redmon and A. Farhadi, "Yolo9000: Better, faster, stronger," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 6517–6525.

[14] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," *Int. J. Comput. Vis.*, vol. 128, pp. 642–656, Mar. 2020.

[15] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[16] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[17] M. Kang, K. Ji, X. Leng, and Z. Lin, "Contextual region-based convolutional neural network with multilayer fusion for SAR ship detection," *Remote Sens.*, vol. 9, no. 8, p. 860, Aug. 2017.

[18] J. Jiao *et al.*, "A densely connected end-to-end neural network for multiscale and multiscene SAR ship detection," *IEEE Access*, vol. 6, pp. 20881–20892, 2018.

[19] X. Zhu, D. Cheng, Z. Zhang, S. Lin, and J. Dai, "An empirical study of spatial attention mechanisms in deep networks," in *Proc. Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 6687–6696.

[20] Z. Cui, X. Wang, N. Liu, Z. Cao, and J. Yang, "Ship detection in large-scale SAR images via spatial shuffle-group enhance attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 379–391, Jan. 2021.

[21] J. Pang, C. Li, J. Shi, Z. Xu, and H. Feng, "Object detection in large-scale remote-sensing images based on time-frequency analysis and feature optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5512–5524, Sep. 2019.

[22] X. Wang, Z. Cui, Z. Cao, and Y. Tian, "Ship detection in large scale SAR images based on bias classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 1263–1266.

[23] X. Niu, D. Yang, K. Yang, H. Pan, Y. Dou, and F. Xia, "Image translation between high-resolution optical and synthetic aperture radar (SAR) data," *Int. J. Remote Sens.*, vol. 42, no. 12, pp. 4758–4784, Jun. 2021.

[24] A. Neubeck and L. Van Gool, "Efficient non-maximum suppression," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, vol. 3, 2006, pp. 850–855.

[25] Y. Wang, C. Wang, H. Zhang, Y. Dong, and S. Wei, "A SAR dataset of ship detection for deep learning under complex backgrounds," *Remote Sens.*, vol. 11, no. 7, p. 765, Mar. 2019.

[26] T. Lin, P. Dollár, R. B. Girshick, K. He, B. Hariharan, and S. J. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Mar. 2017, pp. 936–944.

[27] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 8759–8768.

[28] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[29] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, May 2019, pp. 821–830.

[30] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Mar. 2019, pp. 7355–7364.

[31] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Mar. 2017, pp. 764–773.

[32] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.

[33] J. Wang, K. Chen, S. Yang, C. C. Loy, and D. Lin, "Region proposal by guided anchoring," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 2960–2969.

[34] S. Wei *et al.*, "Precise and robust ship detection for high-resolution SAR imagery based on HR-SDNet," *Remote Sens.*, vol. 12, no. 1, p. 167, Jan. 2020.

[35] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[36] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.

[37] B. Li, Y. Liu, and X. Wang, "Gradient harmonized single-stage detector," in *Proc. Conf. Artif. Intell. (AAAI)*, 2019, pp. 8577–8584.

[38] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

[39] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9756–9765.

[40] X. Zhang, F. Wan, C. Liu, R. Ji, and Q. Ye, "FreeAnchor: Learning to match anchors for visual object detection," in *Proc. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 147–155.

[41] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "FoveaBox: Beyound anchor-based object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 7389–7398, 2020.



Chao Zhang received the M.Sc. degree in advanced computer science from Newcastle University, Newcastle upon Tyne, U.K., in 2019. He is currently pursuing the Ph.D. degree with Harbin Engineering University, Harbin, China.

He is also a Joint Ph.D. Student with the Defense Innovation Institute, Academy of Military Sciences, Beijing, China. His research interests include pattern recognition, vehicle reidentification, and computer vision.



Chule Yang received the B.Eng. degree from Wuhan University, Wuhan, China, in 2014, and the M.Sc. and Ph.D. degrees from Nanyang Technological University, Singapore, in 2015 and 2019, respectively.

He was a Visiting Scholar with INRIA Grenoble-Rhône-Alpes, France, in 2019. He is currently an Assistant Professor with the Defense Innovation Institute (DII), Academy of Military Sciences, Beijing, China. His research interests include autonomous systems, object detection, pattern recognition, semantic perception, and deep situation awareness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.



Kaihui Cheng received the B.Eng. degree from the Zhejiang University of Technology, Hangzhou, China, in 2021. He is currently pursuing the M.S. degree in computer science with the Academy of Military Sciences, Beijing, China.

His research interests include image processing and object detection.



Naiyang Guan received the B.S., M.S., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 2004, 2006, and 2011, respectively.

He is currently an Associate Professor with the Defense Innovation Institute (DII), Academy of Military Sciences, Beijing, China. He has authored or coauthored over 60 research articles on top-tier journals, including the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (T-PAMI), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (T-NNLS), IEEE TRANSACTIONS ON IMAGE PROCESSING (T-IP), and IEEE TRANSACTIONS ON SIGNAL PROCESSING (T-SP), and top-tier conferences, including the IEEE International Conference on Data Mining (ICDM), International Joint Conference on Artificial Intelligence (IJCAI), European Conference on Computer Vision (ECCV), and International Joint Conference on Neural Networks (IJCNN). His research interests include machine learning, computer vision, and data mining.



Hongbin Dong received the Ph.D. degree from the College of computer science and information technology from Beijing Jiaotong University, Beijing, China, in 2006.

He is currently a Professor with the Computer Science and Technology Department, Harbin Engineering University, Harbin, China. His research interests include natural computation, multiagent systems, machine learning, and data mining.



Baosong Deng received the B.S. and Ph.D. degrees in control science and engineering from the National University of Defense Technology, Changsha, China, in 2001 and 2006, respectively.

He is currently a Professor with the Unmanned Systems Research Center, Defense Innovation Institute, Academy of Military Sciences, Beijing, China. His research interests include computer vision, pattern recognition, virtual reality, and application technology of unmanned systems.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:21:01 UTC from IEEE Xplore. Restrictions apply.